

THE ASSOCIATION OF MATHEMATICS TEACHERS OF INDIA

AMTI – NMTC – **25th October, 2025** – SUB-JUNIOR FINAL

Timings: 10am to 1pm

SJ

Instructions:

1. There are 8 questions, answer all questions. Each question carries 10 marks.
2. Elegant and innovative solutions will get extra marks.
3. Diagram and justification should be given wherever necessary.
4. Before start answering, fill in the FACE SLIP completely.
5. Your **Rough Work** should be in the answer sheet itself.

1. x, y, z are three consecutive integers. If $\frac{x^3 + y^3 + z^3}{xyz}$ is also an integer, find all possible values of (x, y, z) .

2. ABC is an equilateral triangle. $AB = 10\text{cm}$. D is a point outside the $\triangle ABC$ such that $BD = DC$ and $\angle BDC = 120^\circ$. Points M, N are on sides AB and AC respectively such that $\angle MDN = 60^\circ$. Find the perimeter of $\triangle AMN$.

3. Find the value of

$$\frac{2025^4 + 2026^4 + 1}{2025^2 + 2026^2 + 1}$$

4. A car, an hour after starting, developed some technical problem, which detains the car an hour, after which it proceeds with three-fifth of its former speed and arrives 3 hours after time to the destination. But had the mechanical problem happened 100km farther on the road, it would have arrived one and half hour sooner than the earlier case. Find the total distance of the journey.

5. $(a_1, a_2, a_3, a_4, a_5, a_6, a_7)$ is a set of seven positive integers. Find the number of such 7-element set of positive integers if $a_1 \times a_2 \times a_3 = 70$, $a_3 \times a_4 \times a_5 = 71$, and $a_5 \times a_6 \times a_7 = 72$.

6. Solve for x :

$$\frac{x}{24} + \frac{x}{104} + \frac{x}{234} + \frac{x}{414} + \frac{x}{644} + \frac{x}{924} + \frac{x}{1254} = 49.$$

7. In a classroom there are 25 students. Their teacher writes 1 at both ends of the blackboard. The first student adds a 2 in the middle between them; each next student adds the sum of each two adjacent numbers already on the blackboard between them. Hence there are numbers 1, 2, 1 after 1st student; 1, 3, 2, 3, 1 after the second student; 1, 4, 3, 5, 2, 5, 3, 4, 1 after the third student and so on. Find the sum of all numbers on the blackboard after the twenty-fifth student.

8. In $\triangle YRL$, $YR = RL$. Equilateral $\triangle YRV$ and $\triangle RLS$ are drawn on the sides YR and RL respectively in the exterior of the $\triangle YRL$. Line YS and line LV intersects in M, then show that $MV = MS$.

-----X-----X-----X-----